WHO PAYS FOR THE KNOWLEDGE MODELS COPY? MODEL COLLAPSE AS ROGERS' PARADOX

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ABSTRACT

Generative models trained on the output of earlier models lose knowledge across generations, rare knowledge first. The remedies that work all bring in information from outside the loop – fresh human data, verifiers, curators – and the theory of model collapse treats that supply as given: a fixed archive, drawn from a fixed distribution. We argue that the supply is the problem, and that cultural evolution has the theory of it. We read model collapse as Rogers' paradox: a population that comes to rely on copying ends up no better off than one that never copied, because copies go stale unless someone keeps learning for themselves. We make the identification quantitative and test it in three steps. First, a one-parameter form of Rogers' model, fitted only to fully synthetic training, predicts how much rare knowledge small language models retain under other mixtures of fresh data, on two corpora. Second, when the world itself changes, the fresh fraction needed to keep up rises with the rate of change, as Rogers' model says it must and as analyses that assume a fixed target cannot. Third, we close the loop: when the people who write a model's training data can ask the model instead of finding things out, the supply of fresh data becomes endogenous. At equilibrium, the model is exactly as reliable as it is costly for people to find out for themselves, the population is no better off than without the model, and every increase in the share of model-written text must be paid for by more people learning for themselves. Users who check what they are told and relearn when it is wrong break the paradox. Finally, we place synthetic-data filters in the taxonomy of social learning strategies: filters that select plausible text are conformist and accelerate collapse, consistent with prior evidence, while only checks that carry new information act as critical learning.

1 Introduction

Training generative models on the output of earlier models degrades them. Across generations, the tails of the learned distribution thin and disappear, rare content is forgotten before common content, and the models' output narrows [Shumailov et al., 2024, Alemohammad et al., 2024, Bertrand et al., 2024, Guo et al., 2024, Briesch et al., 2023]. Model collapse now has a substantial theory: it can be described as a change of scaling laws concentrated in the tails [Dohmatob et al., 2024b], analysed exactly in regression [Dohmatob et al., 2024a], shown to arise from small proportions of synthetic data [Dohmatob et al., 2025], bounded in rate [Suresh et al., 2025, Seddik et al., 2024], and avoided when real data accumulate rather than being replaced [Gerstgrasser et al., 2024, Kazdan et al., 2025]. Its definitions and assumptions are themselves debated [Schaeffer et al., 2026].

The remedies that work share one feature: each brings in information from outside the self-consuming loop. Fresh real data [Alemohammad et al., 2024], retained real data [Gerstgrasser et al., 2024], verifiers of correctness [Feng et al., 2025, Yi et al., 2026], correction toward ground truth [Gillman et al., 2024], and human curation [Ferbach et al., 2024] all import something the model does not already have. And the theory that supports them shares two assumptions. The

outside information is treated as an exogenous archive, there to be drawn on in whatever quantity is needed. And the target distribution is fixed: what the model should know does not change while it is being copied.

Neither assumption holds for the systems that motivate the problem. The fresh text on which future models are trained is written by people, and people increasingly consult the models instead of finding things out for themselves; the supply of fresh data responds to the models it feeds. And the world that text describes keeps changing. Both features are central to an older theory of the same failure. In cultural evolution, Rogers' paradox [Rogers, 1988] is the observation that a population in which copying spreads, because copying is individually cheaper than learning, ends up no better off than a population that never copied: copies go stale, and someone must keep paying to learn. The theory of social learning strategies that grew from it [Boyd and Richerson, 1985, Laland, 2004, Enquist et al., 2007, Rendell et al., 2010] is a theory of exactly the two things collapse analyses take as given: where fresh information comes from, and how fast copied information decays when the world moves.

Here we show that a lineage of models trained on mixtures of synthetic and fresh data follows the dynamics of Rogers' population, and that a one-parameter form of the model predicts how much rare knowledge the lineage retains; a small transformer shows the same pattern (Sections 3–4). We demonstrate that when the world changes, the lineage needs more fresh data to keep up, as Rogers' model requires (Section 5). Closing the loop between models and the people who write their data, we show that the model ends up exactly as reliable as finding out is costly, and that the knowledge models copy must be paid for by people who keep learning for themselves (Section 6). Finally, we show that synthetic-data filters are social learning strategies, and that those which select plausible text act as conformist copying and accelerate collapse (Section 7).

Relation to the closest work. Two recent papers bring social dynamics to AI and collapse. Collins et al. [2026] simulate Rogers' paradox in a population that can also learn from an AI system, which itself updates by tracking the population's mean. They find the paradox intact – at equilibrium, adding the AI leaves collective understanding where individual learning alone puts it – and show that it is broken when people can override the AI by relearning after a failed copy, a form of critical social learning; they also study how often the AI should update and how reliance can erode people's own ability to learn. Their AI thus already learns from the people who learn from it. What they explicitly set aside is the part that defines model collapse: models trained on the output of other models. We take that part as central. Our model has a second copying channel, the share s of model-written text in the training data, and its equilibrium is in closed form, which makes two things visible that the simulations do not state: the model's accuracy at use is pinned at $1 - c$ whatever the model's architecture, and the share of people who must learn for themselves scales as $1/(1 - s)$. Where our results overlap, they agree: the critical asker of Section 6 is their override strategy. And the question they leave open, how an AI changes the rate at which the world changes, is the one our drift experiments begin to quantify from the other side: how the rate of change sets the fresh data a model lineage needs. Baumann et al. [2026] model people choosing between effort and a generative model whose benefit declines with use, and show that welfare can fall below its pre-AI level; their model does not use social-learning theory or the model-to-model loop. Within cultural evolution, Guo et al. [2026] and Ren et al. [2024] read collapse through iterated learning, which explains which structures degrade and in what order. Peterson [2024] argues that reliance on AI narrows public knowledge, and Brinkmann et al. [2023] set out the broader programme of studying machines as participants in cultural evolution. None of these treats the model-to-model loop and the human-to-model loop as one Rogers chain, solves its equilibrium, or tests the chain's quantitative predictions on trained language models; those are our contributions.

2 Background: Rogers' paradox and social learning strategies

In Rogers' model [Rogers, 1988], the best behaviour in an environment changes with probability u per generation. An individual learner pays a cost c and acquires the currently correct behaviour; a social learner copies a random member of the previous generation at no cost, and is right only if its model was right and the environment has not changed since. Social learning spreads while rare, because its copies come mostly from learners, and stops spreading when social and individual learners do equally well. At that point, the population's mean payoff equals that of a population of individual learners alone: copying has been adopted without making anyone better off.

Three refinements matter here. A *critical* social learner copies, checks the copy against the world, and learns for itself if the copy fails; such learners raise the population's payoff above the baseline, because the check imports information [Enquist et al., 2007]. A *conformist* learner copies what is common rather than a random model; conformity is cheap, but it erodes variation, because rare variants are copied less often than they occur [Boyd and Richerson, 1985]. And a *payoff-biased* learner copies whoever does best, which concentrates the population on the current winner [Boyd and Richerson, 1985, Laland, 2004]. The social learning strategies tournament found that successful strategies copy most of the time but copy well [Rendell et al., 2010].

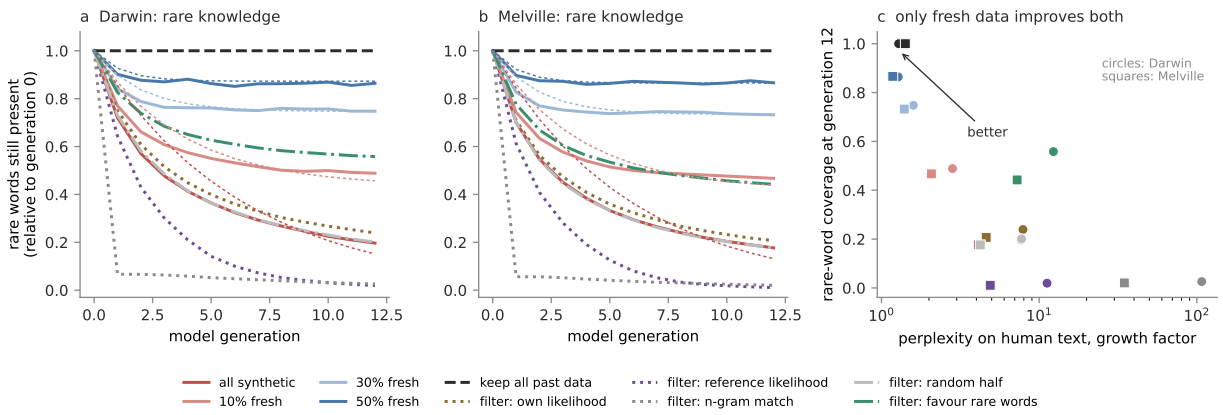


Figure 1: Lineages of language models trained on their own output in a fixed world (word trigram models, 12 generations, mean of 5 seeds). (a, b) Rare-word coverage relative to generation 0. Thin dashed lines: Rogers' chain (Eq. 1) with u fitted to the all-synthetic condition only and applied, unchanged, to the fresh-data conditions. (c) Every condition at generation 12, on coverage and on the growth of perplexity on held-out human text. Filters are discussed in Section 7.

Training a model on another model's output is social learning in the sense of Heyes [2012]: the learner reconstructs competence from another's behaviour through its own learning machinery. The theory of when such learning pays therefore applies to synthetic-data training directly.

3 Collapse as Rogers' chain

Consider a lineage of models. Generation $t+1$ is trained on a corpus of which a fraction ρ is synthetic, sampled from generation t, and $1-\rho$ is fresh. A unit of knowledge carried by the synthetic part survives only if generation t held it and the copy preserved it; let u be the probability per generation that it does not. In Rogers' model u is the rate at which the world changes. In a lineage of models it also includes the chance that a rare item is simply not sampled when the previous model writes its corpus, which is the mechanism of collapse in a fixed world [Shumailov et al., 2024]. The share q_t of knowledge the lineage holds then obeys

$$q_{t+1} = (1-\rho) + \rho\,(1-u)\,q_t, \qquad q^* = \frac{1-\rho}{1-\rho(1-u)}. \tag{1}$$

This is the recursion for the accuracy of copied behaviour in Rogers' population with a fraction ρ of copiers. It predicts that only $\rho = 1$ collapses, at rate $(1-u)^t$; that any fresh fraction buys a plateau q^*; and that knowledge which fresh data rarely covers faces a larger effective ρ and so declines first. Unlike the statistical analyses cited above, Eq. 1 makes no assumption about the model class, and it is deliberately minimal; its value is that a single measured quantity, u, links all mixtures, and that it extends naturally to a world that changes and to a supply of fresh data that responds to the model, the two settings the following sections test.

4 Experiments in a fixed world

Setup. We train lineages of language models on two corpora from Project Gutenberg, Darwin's *On the Origin of Species* and Melville's *Moby-Dick*. Each corpus is split by sentence into a test set (10%), never used for training; a reference set (20%), used only by one filter (Section 7); and a pool (70%) from which the first corpus and all fresh data are drawn. The main model is a word trigram language model interpolating trigram, bigram, and unigram estimates, chosen because its knowledge can be counted exactly; we repeat the key conditions with a small character-level transformer. Generation 0 is trained on 30 000 words from the pool. Each later generation is trained on a new corpus of the same size, built from the previous model's samples according to the condition, for 12 generations and 5 seeds.

Measures. *Rare-word coverage*: of the words outside the pool's 1 000 most frequent, the share still present in the training corpus, relative to generation 0. *Perplexity* on the human test set, as a growth factor over generation 0. *Variety*: distinct three-word sequences per 1 000 words of the model's own samples.

Table 1: Generation 12, mean of 5 seeds. Coverage: rare-word coverage relative to generation 0. Predicted: Rogers' chain with u fitted to the all-synthetic condition of the same corpus ($u = 0.15$ Darwin, 0.16 Melville). Perplexity: growth factor on held-out human text.

	Darwin			Melville		
condition	coverage	predicted	perplexity	coverage	predicted	perplexity
all synthetic	0.20	0.15	7.7	0.18	0.13	4.1
10% fresh	0.49	0.46	2.8	0.47	0.44	2.1
30% fresh	0.75	0.75	1.6	0.73	0.73	1.4
50% fresh	0.86	0.87	1.3	0.87	0.87	1.2
accumulate all past data	1.00	–	1.3	1.00	–	1.4
filter: own likelihood	0.24	–	7.9	0.21	–	4.6
filter: reference likelihood	0.02	–	11.2	0.01	–	4.9
filter: n-gram match	0.03	–	107.5	0.02	–	34.7
filter: random half	0.20	–	7.7	0.18	–	4.2
filter: favour rare words	0.56	–	12.3	0.44	–	7.3

Results. Trained only on its own samples, the trigram model loses four in five of its rare words in twelve generations, and its perplexity on human text grows by a factor of 7.7 on Darwin and 4.1 on Melville (Table 1); a sample from generation 0 reads "i cannot see that a genus become fitted for most justly celebrated elder huber, but in all the large stock", and one from generation 12 "struggle of the on that may inheritable detected i hybrids, kept." Any fresh fraction halts the decline. Fitting u of Eq. 1 to the all-synthetic condition gives 0.15 on Darwin and 0.16 on Melville; with that value and no further fitting, the chain predicts the coverage retained with 10%, 30%, and 50% fresh data to within 0.03 on both corpora (Fig. 1a,b). The chain underestimates the all-synthetic condition at generation 12 because the words that survive longest are the least rare, a heterogeneity that a single u ignores. Keeping every past corpus avoids collapse, as Gerstgrasser et al. [2024] found: in Rogers' terms the original human corpus is a population of individual learners that never leaves.

A neural replication. To check that the pattern is not a property of count-based models, we repeated four conditions with a small character-level transformer (2 layers, width 128, context 128 characters) trained afresh each generation on 60 000 characters of Darwin's text and sampled at temperature 1 (6 generations, 3 seeds; Fig. 2). The qualitative picture is the same. Under fully synthetic training, error on held-out human text rises at every generation (1.77 more bits per character by generation 6), and rare-word coverage falls to 4% of its starting value; with 30% fresh data, both settle within two generations (coverage 0.41 of its starting value at generation 6, compared with 0.04; Welch $p = 0.002$, Cohen's $d = 16.5$). Keeping the half of the sampled lines the model finds most likely is worse than keeping a random half (5.18 against 3.93 bits per character at generation 6, Welch $p = 0.02$, $d = 4.7$, bootstrap 95% CI of the difference $[0.85, 1.52]$), and the random half is indistinguishable from no filter at all (3.93 against 3.86, Welch $p = 0.37$), as it should be. The quantitative fit is looser than for trigrams: a character model sampled at temperature 1 misspells most rare words in its first generation, so the loss is front-loaded rather than geometric, and the one-parameter chain fitted to fully synthetic training under-predicts the fresh-data plateau (0.35 against 0.42). With three seeds per condition the rank-based tests cannot reach $p < 0.05$ (Mann–Whitney $p = 0.08$–0.10 for every comparison); we report them for completeness.

5 A moving world

Analyses of collapse assume a fixed target: the model is judged against the distribution its first generation saw. In Rogers' model the world moves, and the parameter u is precisely the rate at which copied knowledge goes stale because of it. If the identification holds, a lineage of models in a changing world should need more fresh data to keep up, and the extra need should grow with the rate of change.

Setup. We let the world change continuously. The world at generation t is a window of 3 000 consecutive sentences in the concatenated text of Darwin's and Melville's books, which slides forward by a fixed number of sentences per generation, wrapping around; new vocabulary therefore keeps arriving and old vocabulary keeps leaving. Fresh data are drawn from the current window, and every tenth sentence of the window is held out. Each generation is trained on 30 000 words, a fraction f fresh and the rest sampled from the previous model. Coverage is measured against the current window's rare words, relative to a corpus of the same size drawn entirely fresh from the current window, and averaged over generations 6 to 12, after the initial transient (3 seeds).

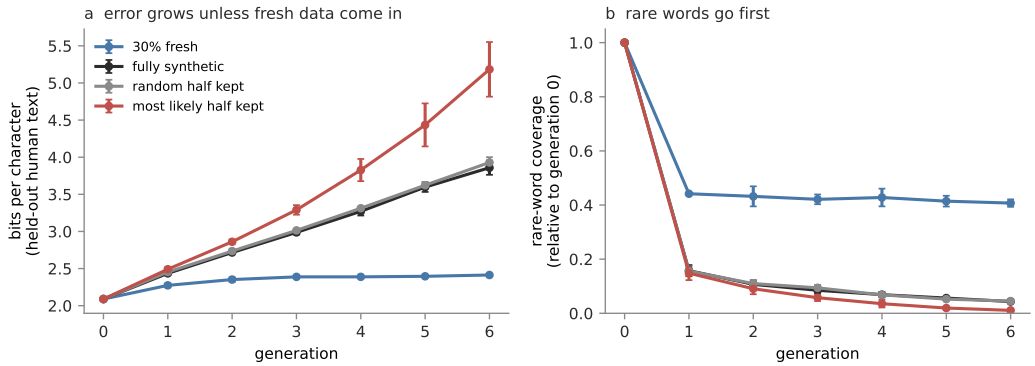


Figure 2: Recursive training of a small character-level transformer (Darwin; 6 generations; mean ± s.d. over 3 seeds). (a) Bits per character on held-out human text. (b) Rare-word coverage of the training corpus, relative to generation 0. Fresh data stop the decline; keeping the most likely half of the samples accelerates it relative to keeping a random half.

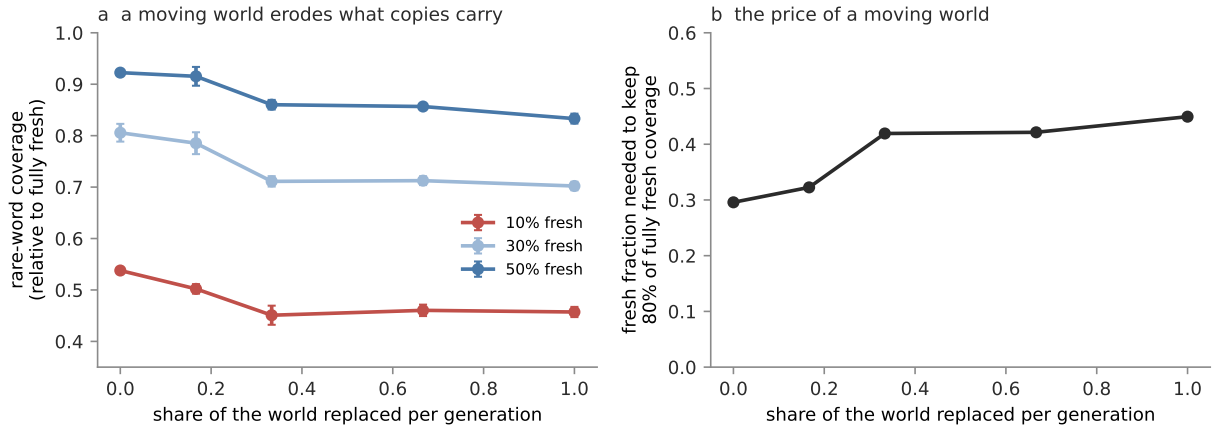


Figure 3: Recursive training in a world that keeps changing (word trigram models; the world is a window of 3 000 sentences sliding through the concatenated corpora; mean of generations 6–12 and 3 seeds; bars: s.d. over seeds). (a) Coverage of the current world's rare words, relative to a fully fresh corpus, against the share of the world replaced per generation. (b) The fresh fraction needed to keep 80% of fully fresh coverage rises from about 0.30 in a static world to about 0.45 when the world is replaced every generation.

Results. At every fresh fraction, coverage falls as the world changes faster (Fig. 3a): with 30% fresh data, from 0.81 in a static world to 0.70 when the window is replaced every generation; with 10%, from 0.54 to 0.46. Equivalently, the fresh fraction needed to retain 80% of what a fully fresh corpus would cover rises from about 0.30 to about 0.45 (Fig. 3b). The effect saturates once about a third of the world is replaced per generation: at that point almost nothing a copy carries about the current world's rare vocabulary is still current, which in Rogers' terms is a staleness probability u at its ceiling. The general point is that the fresh data a lineage needs is not a property of the model or the corpus alone; it depends on how fast the world the corpus describes is changing, and fast-changing domains should be the first to degrade as synthetic text accumulates.

6 Closing the loop: who supplies fresh data?

The analyses above, and the literature, treat fresh data as a resource outside the system. It is not. Fresh text is written by people, and people who can ask a model instead of finding something out will often do so. We now couple the two loops.

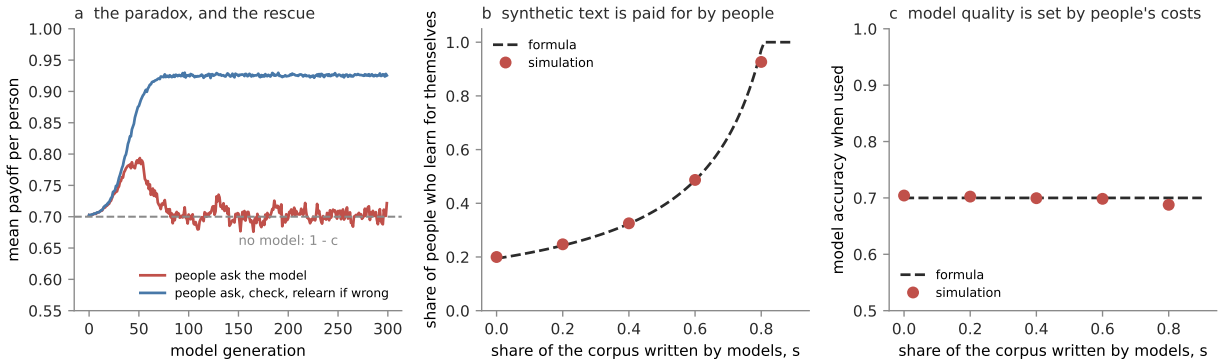


Figure 4: The human-AI knowledge ecosystem, simulated (4 000 facts, 4 000 people, change rate $u = 0.05$, cost of finding out $c = 0.3$, checking cost 0.05). (a) Mean payoff after a model is introduced: when people simply ask, payoff rises and then returns to the no-model baseline $1 - c$; when they check the answer and relearn if it is wrong, it stays well above. (b) Equilibrium share of people who learn for themselves, against the share s of model-written text: simulation and Eq. 2 (corrected for the share of facts written about in each generation). (c) The model's accuracy when used stays at $1 - c$ whatever s.

Model. Knowledge consists of many independent facts, each of which changes to a new value with probability u per generation, where a generation is one model release. A fraction h of people learn each fact they need for themselves, at cost c, and are always right; the rest ask the current model, at negligible cost, and are right with probability a_t, the model's accuracy at the time of use. What people write becomes the training corpus for the next model, together with a share s of text the model writes directly. Learners write what they found; askers write what they were told. People adjust their strategy toward whichever pays more.

Proposition 1. *Let $f = (1 - s)\,h$ be the fresh fraction of the corpus. The model's accuracy at use obeys $a_{t+1} = (1 - u)\,[\,f + (1 - f)\,a_t\,]$. If $0 < h^* < 1$ at equilibrium, then*

$$a^* = 1 - c, \qquad f^* = \frac{u\,(1 - c)}{c\,(1 - u)}, \qquad h^* = \frac{f^*}{1 - s}, \qquad \bar{W}^* = 1 - c, \tag{2}$$

where $\bar{W}^*$ is mean payoff. An interior equilibrium exists only while $s < 1 - f^*$.

Proof. Askers earn a and learners $1 - c$; an interior equilibrium of payoff-driven strategy change requires $a^* = 1 - c$. At the fixed point of the accuracy recursion, $a^* = (1 - u)f/(1 - (1 - u)(1 - f))$; setting it to $1 - c$ and solving for f gives f^*. Mean payoff is $h^*(1 - c) + (1 - h^*)a^* = 1 - c$. □

Proposition 1 has three consequences for machine learning. *First*, at equilibrium the model is exactly as reliable as it is costly for people to find out for themselves. Its accuracy is set not by its capacity or its data volume but by the cost of the alternative it displaces, at least in this model. *Second*, the population is no better off than it would be without the model: this is Rogers' paradox for the human-AI ecosystem. *Third*, text written by models is paid for in human effort. The share of people who must keep learning for themselves rises as $1/(1 - s)$ with the share of model-written text, and beyond $s = 1 - f^*$ the model is no longer worth asking at all. In the model-collapse literature the remedy is "keep fresh data"; Proposition 1 says how much, who supplies it, and what it costs them.

Simulation. An agent-based simulation of the same model, with strategy change by payoff-proportional imitation and a small mutation rate, confirms the equilibrium (Fig. 4). After a model is introduced, mean payoff first rises, while most people still learn and the model is accurate, and then falls back to $1 - c$ as reliance spreads and the model's corpus fills with copies (Fig. 4a). The share of learners matches Eq. 2 across synthetic shares from 0 to 0.6, and the model's accuracy at use stays at $1 - c$ throughout (Fig. 4b,c).

The rescue. If people who ask the model also check its answer against the world, at a cost below that of finding out, and learn for themselves when it is wrong – critical social learning in the sense of Enquist et al. [2007] – their payoff is $1 - c_v - (1 - a)c$, which exceeds the baseline whenever checking is cheaper than the expected cost of being wrong, and what they write back is correct. In the simulation, mean payoff settles at 0.93 against the baseline of 0.70 (Fig. 4a). The implication for the design of AI systems is direct: tools that make it cheap for users to verify an answer against a

source are not a convenience but the mechanism that keeps the ecosystem's knowledge, and the models trained on it, from collapsing to the paradox.

7 Filters are social learning strategies

Most pipelines do not train on raw synthetic data; they filter it. In social-learning terms, a filter is a rule for choosing what to copy, and the theory distinguishes three kinds.

A *conformist* filter prefers what is common. Selecting synthetic samples by their likelihood under the generator, by the perplexity of a reference model, or by overlap with existing text are conformist in this sense: the criterion is resemblance, and rare content is by definition less resembled. Theory predicts that such filters erode variation faster than copying at random [Boyd and Richerson, 1985], and the evidence agrees. In image models, sampling biased toward quality preserves precision while accelerating the loss of diversity [Alemohammad et al., 2024], and selection by typicality precipitates collapse, doing worse than random selection [Qiao et al., 2026]; iterated curation by any fixed score concentrates the model on the highest-scoring samples [Ferbach et al., 2024]. Our text results are consistent (Table 1, Fig. 1c). For the trigram model, filtering by the generator's own likelihood is indistinguishable from discarding at random; for the transformer, whose likelihood ranks whole lines more sharply, it is clearly worse than random and accelerates the loss of rare words (Fig. 2). Filtering by a reference model's likelihood, or by matching human n-grams, loses nearly every rare word within twelve generations, degrades perplexity on human text more than doing nothing, and sharply reduces the variety of the model's own output. A filter that inverts the bias, favouring rare words, preserves more of them but degrades perplexity faster: it keeps rare words by keeping the errors around them.

A *payoff-biased* filter keeps what scores best on a task, such as outputs that a reward model rates highly or that pass a check. It concentrates the population on the current best and, like selection, reduces variation even when it improves average quality; iterated self-improvement narrows output diversity even when it preserves correctness [Briesch et al., 2023, Ferbach et al., 2024], and verifiers can raise a model only to the level of the verifier [Yi et al., 2026].

A *critical* filter tests a copy against the world and replaces it when it fails. Verification against ground truth [Feng et al., 2025], correction toward an expert [Gillman et al., 2024], and training on one's own reasoning only when it reaches a verified answer [Zelikman et al., 2022] belong here, and they are the filters that prevent collapse. The distinction that matters is not how much a filter discards but where its criterion gets its information: from resemblance to what the system already has, or from the world.

Two results complicate the picture and are worth reconciling. Selecting synthetic text with a detector of machine-generated text can prevent collapse [Drayson et al., 2025]: such a detector penalises the over-represented features of model output, which makes it closer to an anti-conformist filter than to the likelihood filters above, and our rare-favouring filter shows the same partial benefit. And selecting synthetic data by n-gram similarity to human text had little effect in a setting that mixes synthetic with human data at the token level [Zhu et al., 2025]; in our fully recursive setting the same criterion, applied every generation, compounds.

8 Discussion

What the identification buys. The statistical theory of collapse is more detailed than Eq. 1 wherever both apply. What the social-learning reading adds is a frame in which the supply of fresh information and the rate of change of the world are variables rather than assumptions, and a taxonomy in which the effect of a filter can be predicted from where its criterion gets its information. These are the levers an engineer of a data pipeline, and a designer of AI products, actually hold.

Predictions. The analysis makes predictions that larger experiments could refute: the knowledge a lineage retains under a given fresh fraction should follow Eq. 1 with a measured u; the fresh fraction needed to keep up should rise with how fast a domain's facts change, which predicts that fast-changing domains degrade first as synthetic text accumulates; likelihood- and typicality-based filters should accelerate the loss of rare knowledge at scale as they do in small models; and, in deployed systems, the share of users who verify answers should govern the long-run reliability of models trained on what users write.

A wider correspondence. The identification is one instance of a broader parallel between the layers of modern training – pretraining, fine-tuning, adaptation in context, and training on other models' output – and the layers of biological adaptation – evolution, individual learning, and social learning [Stephens, 1991, Boyd and Richerson, 1985, Henrich, 2015]. The parallel is not one of mechanism; it is that both are stacks of adaptive processes with different

speeds and costs that must divide the work of keeping up with a changing world, and results that depend only on rates, costs, and channels transfer between them.

Limitations. The language models are small, chosen so that knowledge could be counted exactly, and rare-word coverage is a proxy for knowledge; whether plausibility filters behave the same way under strong reward or quality models at scale is the most important open test. The drifting world is a mixture of two corpora, not a changing world of facts. The ecosystem model treats facts as independent and strategies as population-level averages, ignores that people differ in what learning costs them, and assumes that the next model is trained on what people write; richer versions – heterogeneous costs, several competing models, markets for fresh data – are natural extensions.

9 Conclusion

Read as Rogers' paradox, model collapse is the oldest problem in the theory of social learning, met at industrial scale. Cultural evolution's answer was never to stop copying: it was to keep learners in the population and to check copies against the world. Taking that answer seriously changes the question machine learning asks about collapse. The question is not only how much fresh data a model needs, but who produces it, what it costs them, and whether the systems built on their work leave them any reason to keep producing it.

Code and data. Code for every experiment and figure, and the per-seed results, are available at `https://github.com/drnamlabs-gif/rogers-model-collapse`.

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